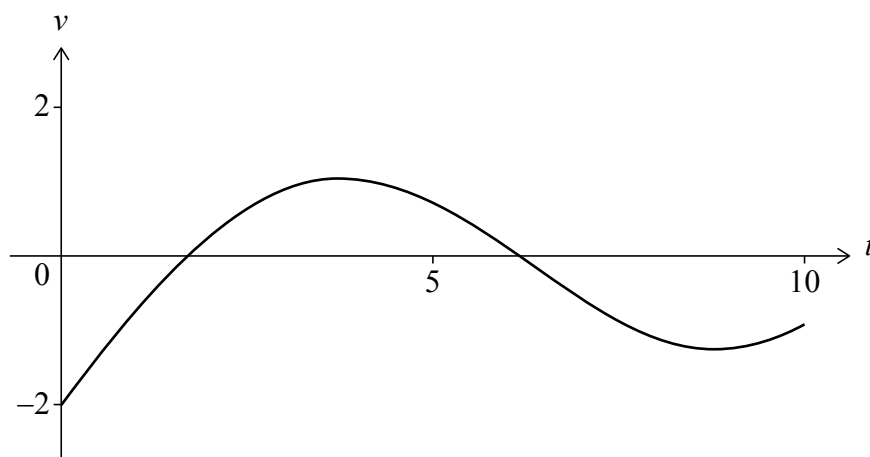


4. [Maximum mark: 6]

A particle moves in a straight line such that it passes through a fixed point O at time $t = 0$, where t represents time measured in seconds after passing O. For $0 \leq t \leq 10$ its velocity, v metres per second, is given by $v = 2 \sin(0.5t) + 0.3t - 2$.

The graph of v is shown in the following diagram.



- (a) Find the smallest value of t when the particle changes direction. [2]

The displacement of the particle is measured in metres from O.

- (b) Find the range of values of t for which the displacement of the particle is increasing. [2]
 (c) Find the displacement of the particle relative to O when $t = 10$. [2]

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